

Running the first ENM/SDM

ENM

Data

```
library(mgcv)
library(gratia)

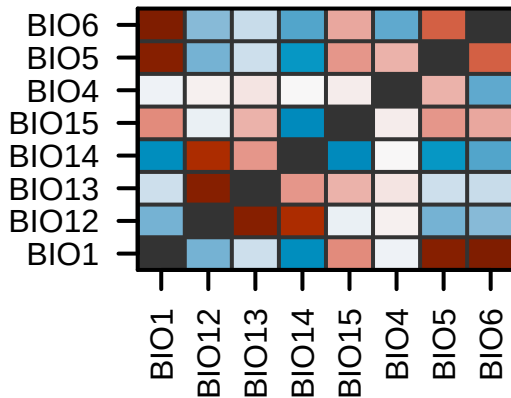
d <- read.csv("../data/occurrences.csv")
head(d, n = 3)
```

	wc2.1_10m_bio_1	wc2.1_10m_bio_12	wc2.1_10m_bio_13	wc2.1_10m_bio_14
1	14.709969	510	60	20
2	11.454646	775	89	44
3	8.854438	933	106	60

	wc2.1_10m_bio_15	wc2.1_10m_bio_4	wc2.1_10m_bio_5	wc2.1_10m_bio_6	cell
1	25.68323	686.3660	32.04650	2.11375	620998
2	21.44033	589.9808	25.82825	-0.33375	621014
3	18.08941	578.0707	22.92975	-2.12900	618853

	x	y	occ
1	-0.4166667	42.08333	1
2	2.2500000	42.08333	1

Correlation among variables



- ▶ Bioclimatic variables are highly correlated with each others.
- ▶ A subset of them should be used for ENM/SDM.

Selecting variables

Variable selection

- ▶ The usual statistical tricks ...
- ▶ Informed by the **biology** of the species.

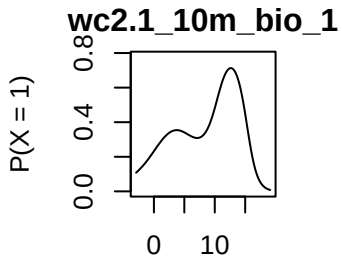
Because I do not know the biology of the species, I will use a statistical trick: Akaike information criterion (AIC).

Full ENM

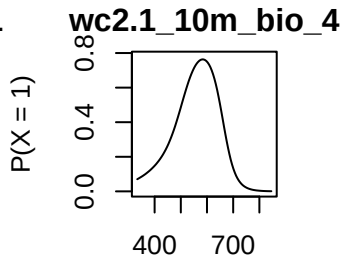
```
enm <- gam(  
  occ ~ s(wc2.1_10m_bio_1, k = 5) +  
    s(wc2.1_10m_bio_4, k = 5) +  
    s(wc2.1_10m_bio_12, k = 20) +  
    s(wc2.1_10m_bio_15, k = 5),  
  data = d,  
  family = "binomial"  
)  
k.check(enm)
```

	k'	edf	k-index	p-value
s(wc2.1_10m_bio_1)	4	3.967871	0.9937998	0.3825
s(wc2.1_10m_bio_4)	4	3.739801	0.9914376	0.3250
s(wc2.1_10m_bio_12)	19	11.160664	0.7681976	0.0000
s(wc2.1_10m_bio_15)	4	3.846506	0.9902516	0.2575

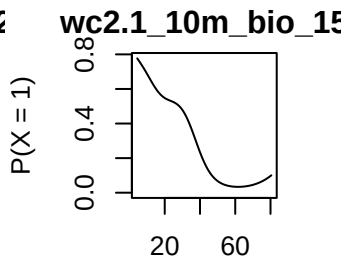
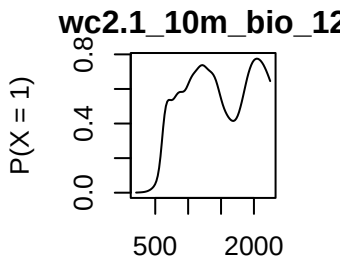
Inspect ENM



climate



climate



SDM

Prepare rasters

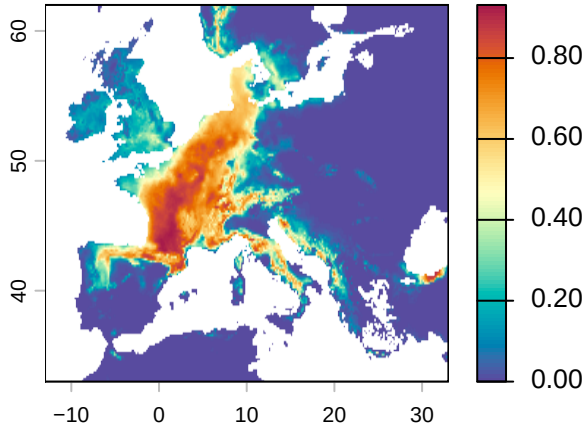
Load layers.

```
library(terra)

ff <- paste0("../data/", unlist(smooth_terms(enm)), ".tif")
r <- rast(ff)
roi <- ext(-13, 33, 33, 62) # roi of Europe
r <- crop(r, roi) # crop to Europe
```

Project distribution

```
sdm <- predict(r, enm, type = "response")  
  
plot(sdm, col = hcl.colors(100, "Spectral", rev = TRUE))
```



Binary projection

If we are interested in a binary map, e.g. showing the climatic range of the species, we need to binarize this continuous value into 0/1.

True skill statistics (TSS).

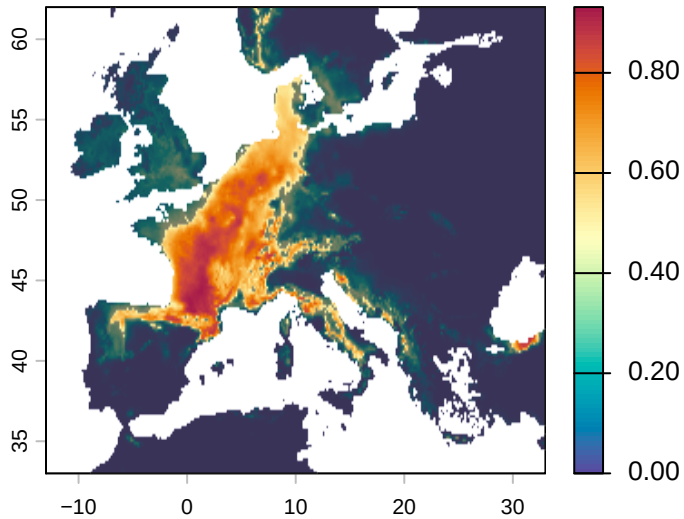
1. Pick a threshold value th .
2. Set all cells $< th$ to 0.
3. Set all cells $\geq th$ to 1.
4. Calculate the TSS.
5. Repeat for a different value of th .
6. Report maximum TSS and use th for which $TSS = \max(TSS)$.

$$TSS = \frac{TP \times TN - FP \times FN}{(TP + FN)(TN + FP)}$$

Binary projection

```
suit <- extract(sdm, d[, c("x", "y")], ID = FALSE)[, 1]
threshold <- seq(0.1, 0.9, by = 0.001)
tss <- rep(NA, length(threshold)) # empty vector for storage
for (i in seq_along(threshold)) {
  p <- ifelse(suit > threshold[i], 1, 0)
  TP <- sum(p == 1 & d$occ == 1)
  FP <- sum(p == 1 & d$occ == 0)
  FN <- sum(p == 0 & d$occ == 1)
  TN <- sum(p == 0 & d$occ == 0)
  tss[i] <- TP / (TP + FN) + TN / (TN + FP) - 1
}
th <- threshold[which.max(tss)] # best threshold
sdm_bin <- ifel(sdm >= th, 1, 0) # binarize map
```

Binary projection



Binary projection

Our map is quite *incorrect* for this species:

<https://www.iucnredlist.org/species/61550/12514105>

Why do we get such bad projections compared to the known range from IUCN?

We will answer this in a next lecture.

Hint: geographic biases in the detections

